YUROU LIU

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EDUCATION

YALE UNIVERSITY — New Haven, CT

B.S. in Physics (Intensive) — Aug 2022 - May 2026 (anticipated)
B.S. in Computer Science
GPA: 3.99/4.00 Overall; 4.00/4.00 in Physics (Intensive)
Relevant Courses: Exoplanets, Many-Body Dynamics, Observational Astronomy (incl. telescope proposal), Cosmology, Research Methods in Astrophysics (incl. telescope proposal), Machine Learning, Classical Mechanics, Electromagnetism, Quantum Mechanics I & II, Statistical Mechanics, Advanced Physics Laboratory, Algorithms, Systems Programming, Computer Vision, Computer Graphics

RESEARCH EXPERIENCE

INDIANA UNIVERSITY BLOOMINGTON — Bloomington, IN
Alexander P. Hixon Summer Research Fellow, Department of Astronomy — Jun 2025 - Present
Mentored by Prof. Cristobal Petrovich and Dr. Hareesh Bhaskar

- Developed novel secular integration codes in C to study secular chaos in hierarchical “3+1” quadruple exoplanet systems (planet orbiting a star, perturbed by a nearby substellar/stellar companion and a fourth distant stellar companion)
- Created surfaces of section from rotating frame Hamiltonian of the “3+1” quadruple system to analytically probe parameter space of chaos
- Demonstrated that near-coplanar and circular “3+1” quadruple systems can excite planetary eccentricities to levels sufficient for high-eccentricity migration
- Manuscript submitted to AAS Journals and currently under review.

CALIFORNIA INSTITUTE OF TECHNOLOGY — Pasadena, CA
Summer Undergraduate Research Fellow, Department of Astronomy — Jun 2024 - Present
Mentored by Dr. Yapeng Zhang and supervised by Prof. Dimitri Mawet

- Performed atmospheric retrievals on CRIRES+ spectra of directly imaged gas giant 2M0249-0557 c and two benchmark brown dwarfs in the β Pictoris moving group
- Developed a nested sampling retrieval code using petitRADTRANS and PyMultiNest to constrain C/O ratio, metallicity, and $^{12}CO/^{13}CO$ isotopologue ratio
- Interpreted retrieved compositions in the context of formation pathways, supporting a star-like formation mechanism for 2M0249-0557 c
- Manuscript submitted to AAS Journals and currently under review.

YALE UNIVERSITY — New Haven, CT
Undergraduate Researcher, Department of Astronomy — Jun 2023 – Present
Mentored by Prof. Malena Rice and Dr. Tiger Lu

- Ran N-body simulations with REBOUND and REBOUNDx to model mirrored ZLK migration of two hot Jupiters in binary star systems
- Estimated occurrence rates of double hot Jupiters using the observed distribution of Gaia binaries hosting hot Jupiters
- Quantified the impact of mass and inclination asymmetries on double hot Jupiter formation
- Results published in the Astrophysical Journal; follow-up paper examining a case study of the WASP-94 system submitted to ApJL and currently under review

AWARDS AND FELLOWSHIPS

Alexander P. Hixon Fellowship, Yale University, 2025
Summer Undergraduate Research Fellowship, California Institute of Technology, 2024
First-Year Summer Research Fellowship in the Sciences & Engineering, Yale University, 2023

PUBLICATIONS

Liu, Y., Bhaskar, H. and Petrovich, C. High-Eccentricity Tidal Migration Driven by Secular Chaos in Wide-Binary Systems. Submitted to AAS Journals (under review)
Liu, Y., Zhang, Y., Xuan, J., Mawet, D., Snellen, I., Stolker, T., Kesseli, A. and Rice, M. Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pictoris Young Moving Group. Submitted to AAS Journals (under review)
Liu, Y., Lu, T. and Rice, M., 2026. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94. *ApJL* 997, L41
Liu, Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems through von Zeipel–Lidov–Kozai Migration. *ApJ* 986, 103
Lu, T., Tajer, H., Hernandez, D.M., Rein, H., **Liu, Y.** and Rice, M., 2025. Collisional Fragmentation Support in TRACE. *RNAAS* 9, 110

ORAL PRESENTATIONS (*INVITED)

247th American Astronomical Society Meeting (Anticipated) — Phoenix, AZ, USA
The Formation of Double Hot Jupiter Systems Through ZLK Migration: WASP-94 — Jan 2026
Princeton Exoplanet Lunch* — Princeton, NJ, USA
Gas Giants in Hierarchical Systems: from Dynamics to Atmospheres — Sep 2025
Emerging Researchers in Exoplanet Science Symposium X — Princeton, NJ, USA
Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group — Jun 2025
246th American Astronomical Society Meeting — Anchorage, AK, USA
Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group — Jun 2025
Boston Area Planetary Science Meeting — Boston, MA, USA
Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group — May 2025
Caltech Student-Faculty Programs Seminar Day — Pasadena, CA, USA
Chemistry and Isotope Ratios of the Directly Imaged Gas Giant 2MJ0249-0557c — May 2025
The 55th Meeting of the Division of Dynamical Astronomy — Toronto, Canada
The Formation of Double Hot Jupiter Systems Through ZLK Migration — May 2024
Yale Astronomy Department Summer Research Symposium — New Haven, CT, USA
The Dynamical History of the Two Hot Jupiters in the WASP-94 Binary Star System — Jul 2023

POSTERS

New York Area Exoplanets Meeting — New York, NY, USA
Chemistry and Isotope Ratios of Substellar Atmospheres in the β Pic Moving Group — May 2025

OBSERVING EXPERIENCE

Palomar Observatory (DBSP): 1 night
Spectroscopy of a Gas Giant Exoplanet in Optical Wavelength

TECHNICAL SKILLS

Languages: Python, C/C++, D, Rust, R

Scientific/Graphics: REBOUND, REBOUNDx, KozaiPy, petitRADTRANS, PyMultiNest, TensorFlow, PyTorch, matplotlib, NumPy, SciPy, Astropy, pandas, GSL, Mathematica, OpenGL, SDL
Tools: Git, Bash, LaTeX, SQL, Slurm

PROJECTS

Generate Fake Transits

- Trained a conditional variational autoencoder on TESS data to generate synthetic exoplanet transits based on input parameters.

L4 and L5 Lagrange Points Visualization

- Visualized particle dynamics near Lagrange points in a star-planet system using OpenGL instanced rendering.

More projects at: yurouninaliu.github.io/projects

TEACHING EXPERIENCE

YALE UNIVERSITY — New Haven, CT
Undergraduate Learning Assistant, Department of Physics — Sep 2023 – May 2024
PHYS 261: Intensive Introductory Physics II
PHYS 260: Intensive Introductory Physics I

OUTREACH TALKS

Yale Undergraduate Kick-off Event — New Haven, CT, USA
Hot Jupiter Formation through Secular Chaos in "3+1" 4-body Systems — Sep 2025

SCIENCE COMMUNICATION AND OUTREACH

Contributor and layout editor, Yale Scientific Magazine

- Created original artwork and designed print layouts
- Provided individualized feedback to contributors
- Improved layout designs for publication
- Led layout workshops to train new designers
- Volunteered at the Resonance Conference to facilitate hands-on STEM classes for local high school students

Workshop leader, Yale Guild of Book Makers

- Taught hands-on workshops in bookbinding

MEDIA

Yale News, "New study offers a double dose of 'hot Jupiters'" — Jun 2025
StarXiv, "Episode 11: Whirling planes, wandering black holes and alien supernovae" — May 2025